

8.7 hours



628 ms

~50,000 x faster

Same algorithm. Same hardware.

Python loops replaced by vectorised NumPy.

Ant Colony System for edge detection | PPGCC / UNESP

Ant Colony System

for image edge detection

1

Ants start at random pixels

Walk through 8-neighbours guided by intensity gradient; sharper edges attract more ants.

2

Each step deposits pheromone

Local update evaporates then redeposits based on gradient; prevents all ants converging to a single path.

3

Threshold the pheromone matrix

After N iterations, Otsu's method on the pheromone map produces the final binary edge image.

How the pheromone works

LOCAL UPDATE every ant step

After each move, the visited pixel is partially evaporated, then redeposited based on local gradient. This prevents all ants from locking onto the same path.

GLOBAL UPDATE after each iteration

Pixels many ants visited get a strong reinforcement, weighted by gradient strength at that pixel. Unvisited pixels continue fading toward zero.

q0 exploitation vs exploration

$q \leq q_0$: ant picks the best neighbour greedily.
 $q > q_0$: ant samples probabilistically from the neighbourhood.
Low q_0 : diffuse edges. High q_0 : sharp, concentrated paths.

Same input — three configurations

input / legacy / new default / tuned

Input

Legacy
K=512 q0=0.5

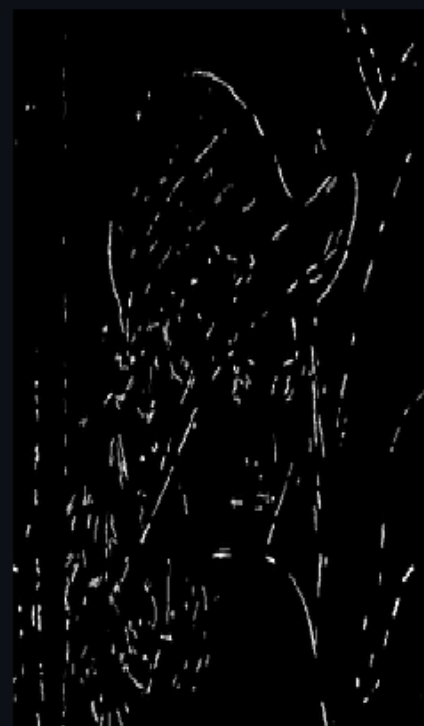
New default
K=2048

Tuned
K=1024 q0=0.7

pikachu
50x43



lena
512x512



Benchmark

~320x

pikachu 50x43

18.9 s -> 59 ms

~50,000x

lena 512x512

~8.7 h -> 628 ms

Image	Original	New (CPU)	Speedup
pikachu 50x43	18.9 s *	59 ms	~320x
lena 512x512	~8.7 h **	628 ms	~50,000x

* measured on this hardware (Python 3.13, Ryzen 7 4800H)

** extrapolated via $O(H \times W \times K \times \text{histLen} \times N)$

GPU available (CuPy / CUDA)

CPU wins for these image sizes on GTX 1650 Ti.

Kernel-launch overhead dominates below a few thousand pixels.

Segmentation quality

F1 score vs Canny edges (proxy ground truth)

Image	Legacy	New default	Tuned	vs legacy
pikachu	57.9%	59.4%	61.7%	+6.6 pp
lena	22.0%	37.4%	27.1%	+70%

For lena, the new default (K=2048) maximises F1 through better recall.

The tuned variant (K=1024, q0=0.7) maximises precision at the cost of recall.

Code, benchmarks, and full gridsearch results:

github.com/yingyangtongxue/ACS-Edge

Open to roles in software engineering, computer vision, or HPC.

#ComputerVision #Python #HPC #SwarmIntelligence #OpenSource